

RUPESH SINGH KARKI

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WORK EXPERIENCE

SociolinQ | Full Stack Developer Intern

Aug 2025 – Present

- Architected a **role-based access control (RBAC)** system and **multi-level approval workflows**, enabling secure and structured data operations across organizational hierarchies.
- Designed and implemented **approval-driven data workflows** ensuring that all critical information changes pass through designated approvers, maintaining **data integrity** and enforcing secure modifications in controlled environments.
- Leveraged **Redis** for caching and state management, reducing redundant database queries and improving response times for frequently accessed endpoints.
- Optimized **PostgreSQL** performance through advanced indexing strategies and query tuning, achieving **20–30% faster data retrieval** across critical modules.
- Built a reusable **activity logging system** using decorators to track endpoint-level changes, ensuring auditability, traceability, and easier debugging in production.
- Developed a fault-tolerant **S3 transaction handler** with rollback mechanisms, maintaining data consistency during partial failures in file upload workflows.
- Applied backend optimization techniques such as **request batching**, **caching strategies**, and efficient schema design to improve system scalability and reliability.

PROJECTS

HearU

[GitHub](#) [Live](#)

- AI-powered confidential mental wellness companion for Indian youth to write journals and seek guidance, featuring React, Redux, TailwindCSS, and ShadCN on the frontend.
- FastAPI backend with JWT authentication, PostgreSQL database, and Gemini generative AI integration for intelligent support and voice assistant features.
- Continuous integration and deployment via GitHub Actions, hosted on Google Cloud Compute Engine with Nginx reverse proxy for scalable, secure delivery.

SamanSetu: Inventory and Storage Management Web App

[GitHub](#) [Live](#)

- Built a full-stack inventory and storage management platform using **React.js** and **FastAPI**, enabling secure authentication, seamless buyer-owner interactions, and scalable containerized deployment.

PUBLICATIONS

A Comprehensive Survey in ANN-based Customer Churn Prediction, In ICDAM-2025 ([Link](#))

EDUCATION

2023 – present	Maharaja Agrasen Institute of Technology (MAIT)	B.Tech in Computer Science and Technology
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SKILLS

Programming	C/C++, Python, TypeScript, JavaScript, Golang
Libraries	TensorFlow, Scikit-learn, FastAPI, React.js, Express.js, Chi
Databases	PostgreSQL, MongoDB, Milvus
Tools	Docker, Redis, RabbitMQ, Celery, Redux

CERTIFICATIONS

HackerRank Problem Solving (Basic) ([Certificate](#))